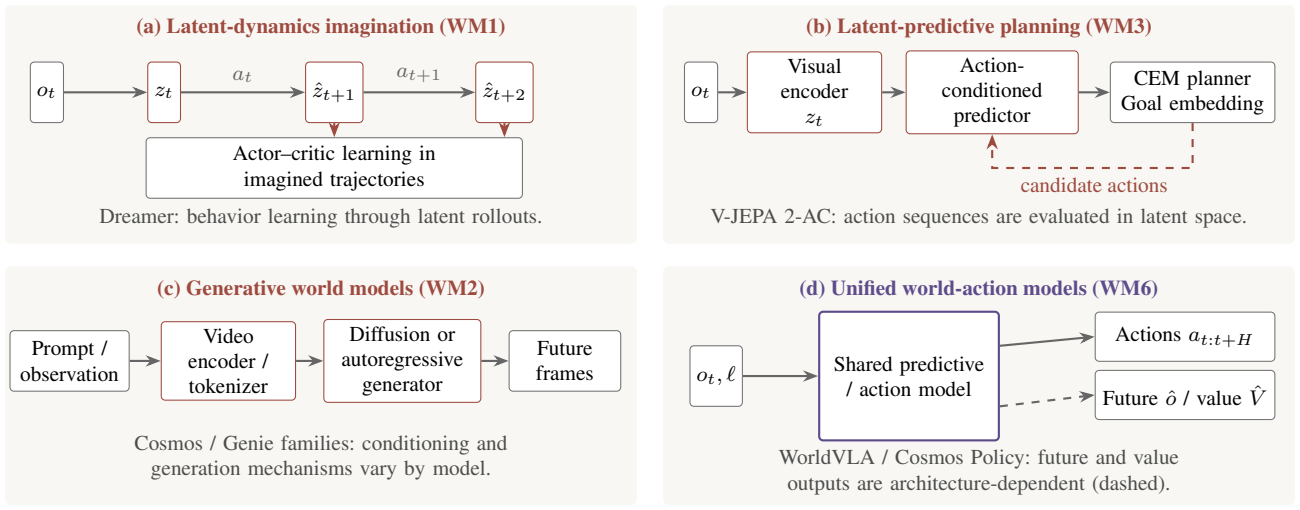

(a) Latent-dynamics imagination (WM1)
o_t
z_t
a_t
$\hat{z}_{t+1}$
a_{t+1}
$\hat{z}_{t+2}$
Actor–critic learning in imagined trajectories
Dreamer: behavior learning through latent rollouts.
(b) Latent-predictive planning (WM3)
o_t
Visual encoder z_t
Action-conditioned predictor
CEM planner Goal embedding
candidate actions
V-JEPA 2-AC: action sequences are evaluated in latent space.
(c) Generative world models (WM2)
Prompt / observation
Video encoder / tokenizer
Diffusion or autoregressive generator
Future frames
Cosmos / Genie families: conditioning and generation mechanisms vary by model.
(d) Unified world-action models (WM6)
o_t, ℓ
Shared predictive / action model
Actions $a_{t:t+H}$
Future $\hat{o}$ / value $\hat{V}$
WorldVLA / Cosmos Policy: future and value outputs are architecture-dependent (dashed).